

3RD HARSH-ENVIRONMENT MASS SPECTROMETRY WORKSHOP

Mass SURFER Field Mass Spectrometer System for Deep Ocean and Planetary Lander Applications

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ABSTRACT

We are currently developing two low-power, miniature mass spectrometer-based field-sampling systems-- a "standard" and CE (capillary electrophoresis)-enhanced version-- called Mass SURFER (Mass Spectrometer Using Rotating Fields for Exploratory Research). The units incorporate the Rotating Field Mass Spectrometer (RFMS) developed at JPL. The current marine prototypes are housed within a 6.5-in. OD pressure vessel, 5.5 feet long that are capable of 2000+ m depth deployments. For planetary applications, we can greatly reduce the size and weight of these vessels. The system components comprise an aquatic-based sensor head, vacuum system, pressure case and associated electronics. Natural fluids, including seawater and brines, with their load of dissolved gases can be directly injected into a submersible vacuum chamber via a nano-electrospray interface (ESI). The ESI is an on-line capillary nozzle capable of high-sensitivity mass spectrometry at nanoliter per minute flow rates. Mass SURFER vacuum levels need only be at the milli-torr level for a quality measurement, and the complete system draws <10 watts when active. The RFMS mass resolution of 1 part in 1000 is comparable with the best of other small mass spectrometers. Its extremely large analytical mass range (from 1 to >100,000 amu) coupled with "soft" ionization techniques make it capable of analyzing practically anything from H₂ and He gas to large dissolved organic compounds such as proteins, peptides and DNA. The Mass SURFER's high sensitivity and low power requirements allow remote, automated, in situ monitoring of dissolved gases, solutes and microbial populations via protein profiling. Future plans, which include analysis of aqueous volatiles and large dissolved organic compounds in deep-ocean hydrothermal vents and cold seeps, as well as NASA projects on the Moon, Mars, and the Jovian moons, especially targeting Europa, will be presented. This work is funded by the Defense Advanced Research Projects Agency (DARPA) via the State of Hawaii's National Defense Center of Excellence for Research in Ocean Sciences (CEROS).